Systems Analysis and Control

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Lecture 20: Drawing Bode Plots, Part 2

Overview

In this Lecture, you will learn:

Simple Plots

- Real Zeros
- Real Poles
- Complex Zeros
- Complex Poles

Review

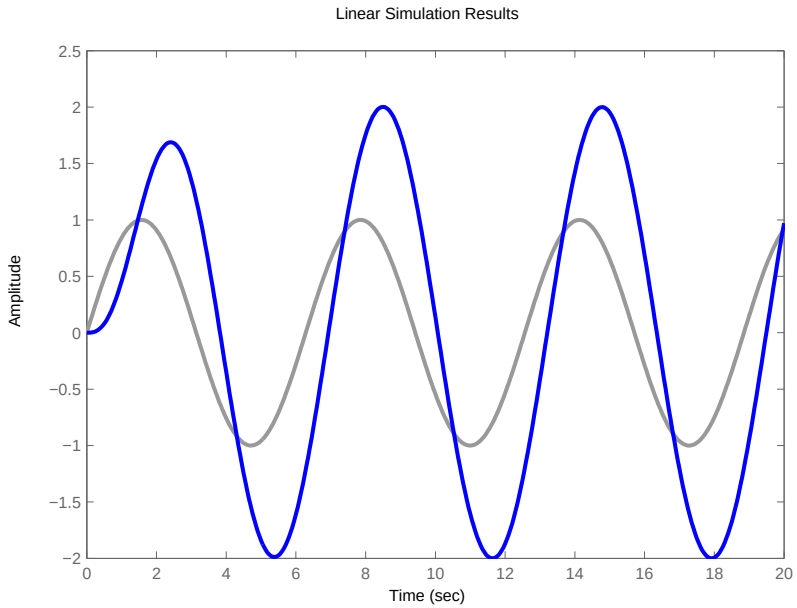
Recall: Frequency Response

Input:

$$u(t) = M\sin(\omega t + \phi)$$

Output: Magnitude and Phase Shift

$$y(t) = |G(\imath\omega)|M\sin(\omega t + \phi + \angle G(\imath\omega))$$



Frequency Response to $\sin\omega t$ is given by $G(\imath\omega)$

Review

Recall: Bode Plot

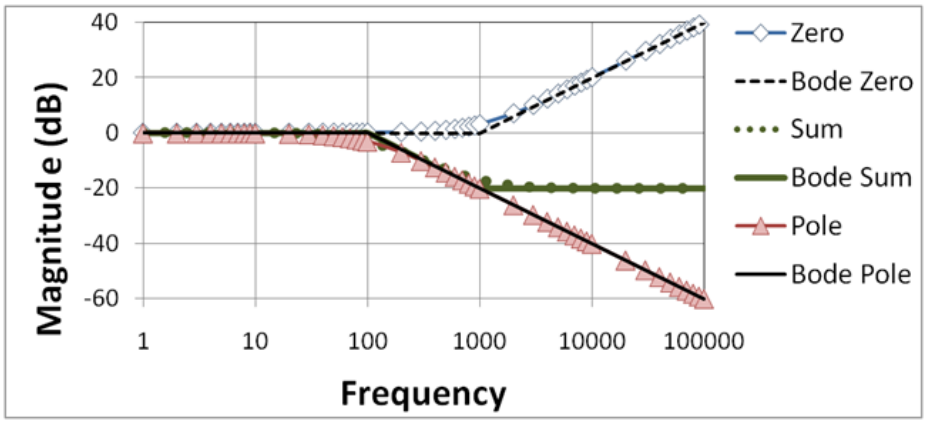
Definition 1.

The Bode Plot is a pair of log-log and semi-log plots:

1. Magnitude Plot: $20 \log_{10} |G(\imath\omega)|$ vs. $\log_{10} \omega$
2. Phase Plot: $\angle G(\imath\omega)$ vs. $\log_{10} \omega$

Bite-Size Chucks:

$$\angle G(\imath\omega) = \sum_i \angle(\imath\omega\tau_{zi} + 1) - \sum_i \angle(\imath\omega\tau_{pi} + 1)$$



$$20 \log |G(\imath\omega)| = \sum_i 20 \log |\imath\omega\tau_{zi} + 1| - \sum_i 20 \log |\imath\omega\tau_{pi} + 1|$$

Recall: Gain Plot of Real Zero: $\angle G(s) = \tau s + 1$

Summary

$$G_1(s) = (\tau s + 1)$$

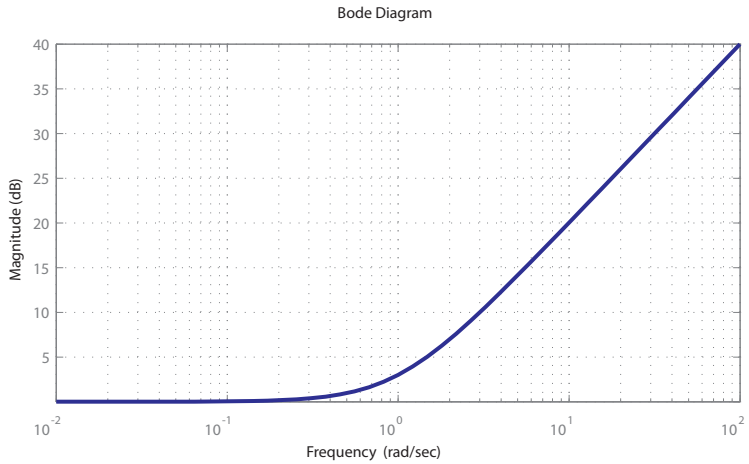
$$|\imath\omega\tau + 1| \cong \begin{cases} 1 & \omega << \frac{1}{\tau} \\ |\imath\omega\tau| & \omega >> \frac{1}{\tau} \end{cases}$$

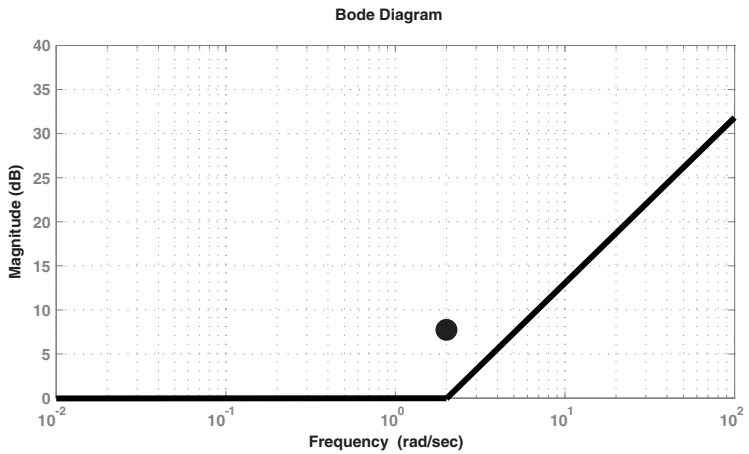
Behaves as

- A constant at low ω.
- A zero at high ω.

Break Point at $\omega = \frac{1}{\tau}$.
Magnitude:

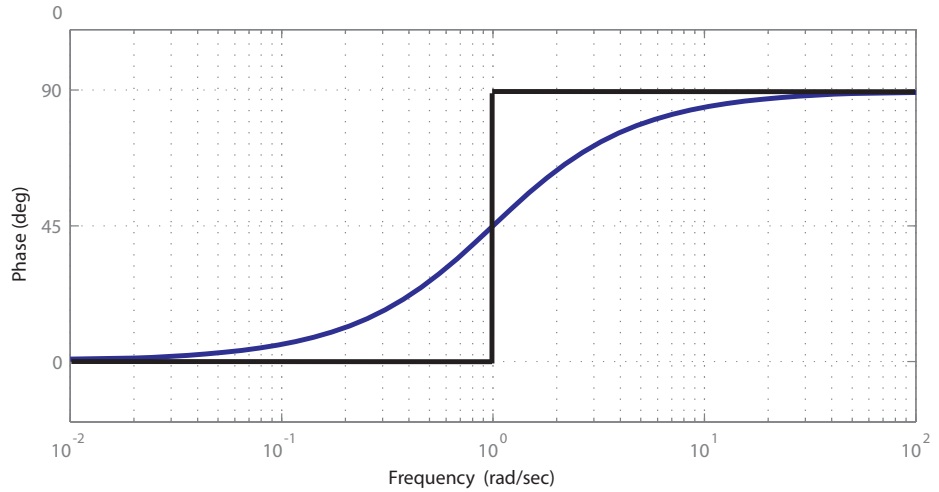
$$20\log|\imath\omega\tau + 1| = \begin{cases} 0 \text{ dB} & \omega << \frac{1}{\tau} \\ 3.01 \text{ dB} & \omega = \frac{1}{\tau} \\ 20\log\omega & \omega >> \frac{1}{\tau} \end{cases}$$





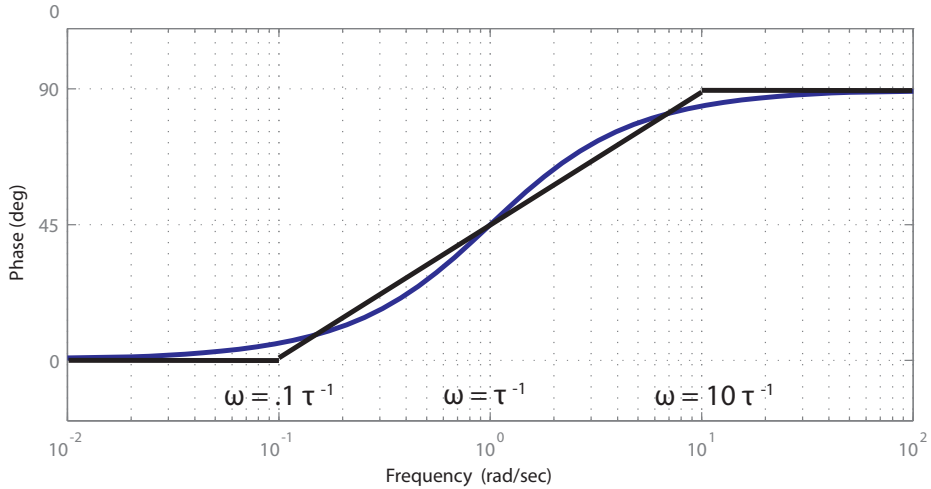
Recall: Phase Plot of Real Zero: $G_0(s) = \tau s + 1$

Inflection points at $\omega = \frac{1}{10}\frac{1}{\tau}$ and $\omega = 10\frac{1}{\tau}$



We can improve our sketch using a line

$$\angle(\imath\omega\tau + 1)$$
$$= \begin{cases} 0^\circ & \omega < \frac{.1}{\tau} \\ 45^\circ\,(1 + \log\omega\tau) & \frac{.1}{\tau} < \omega < \frac{10}{\tau} \\ 90^\circ & \omega > \frac{10}{\tau} \end{cases}$$



Remember: $\frac{1}{\tau} = z$ – Yes, the τ is unnecessarily complicated.

Step 5: Plot the Real Poles

How to Bode Plot Real Poles? Just Flip the Zero Plots

Poles are the opposite of zeros

$$G_{zero}(\imath\omega) = \imath\omega\tau + 1 \qquad G_{pole}(\imath\omega) = \frac{1}{\imath\omega\tau + 1} \qquad G_{zero} = G_{pole}^{-1}$$

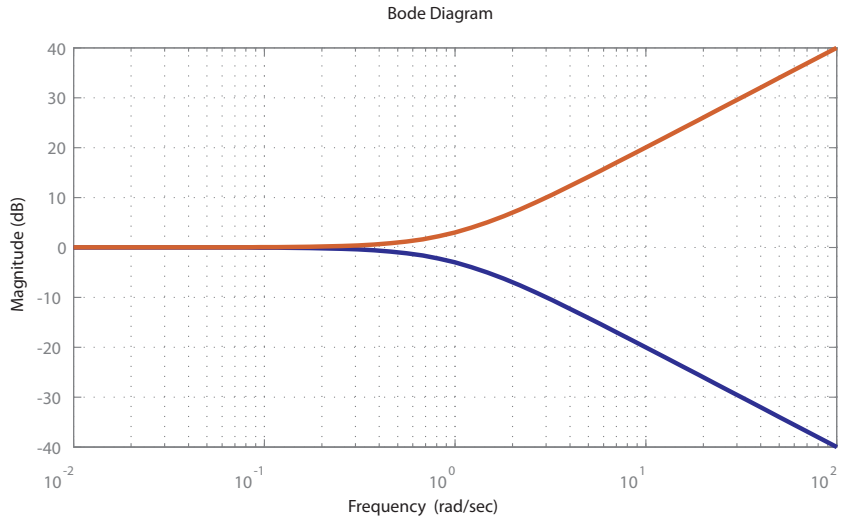
Magnitude:

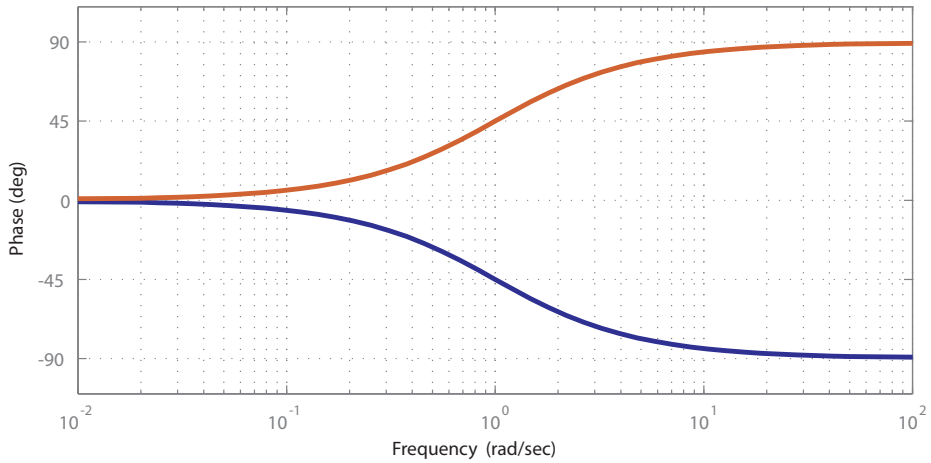
$$|G_{pole}(\imath\omega)| = \frac{1}{|\imath\omega\tau + 1|} = \frac{1}{|G_{zero}(\imath\omega)|}$$

$$20\log|G_{pole}(\imath\omega)| = -20\log|G_{zero}(\imath\omega)|$$

Phase:

$$\begin{aligned}\angle G_{pole}(\imath\omega) &= -\tan^{-1}\omega\tau \\ &= -\angle G_{zero}(\imath\omega)\end{aligned}$$



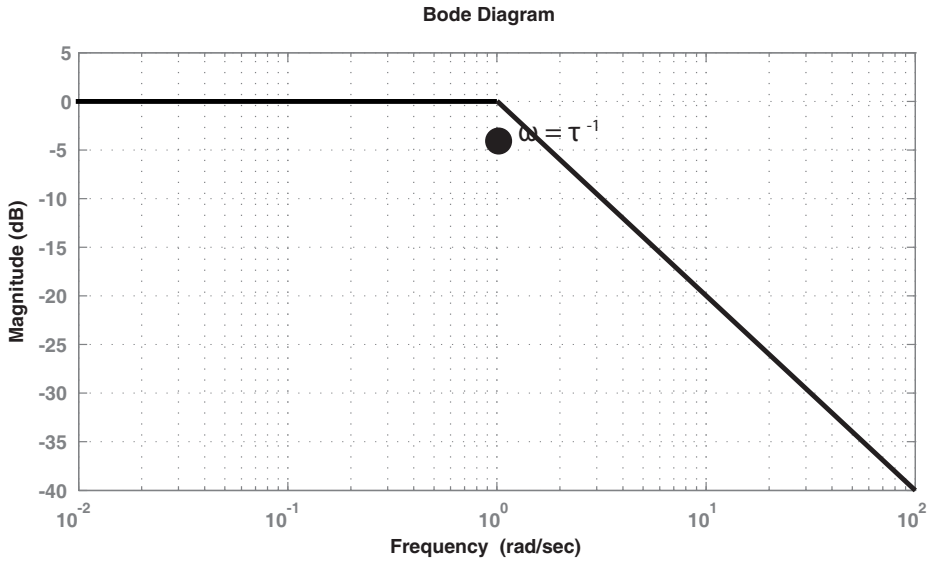


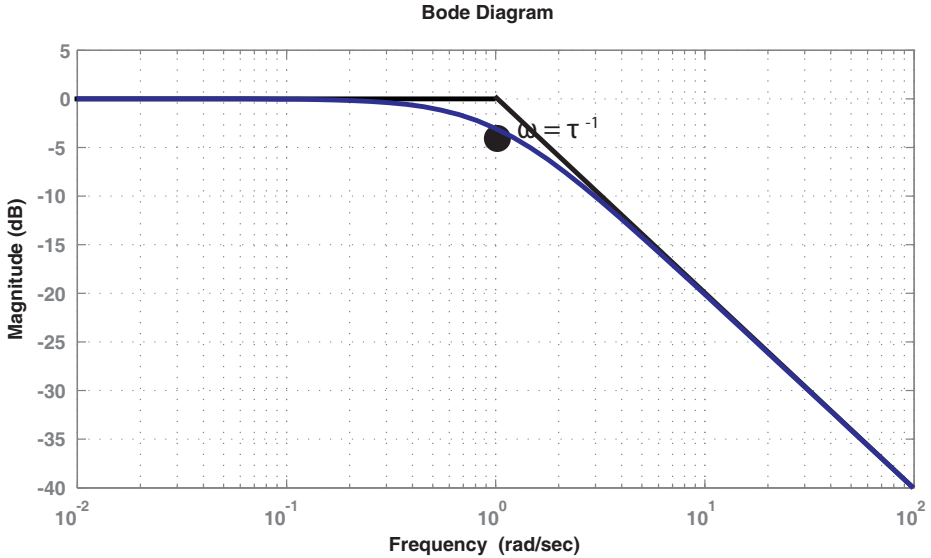
Exact reflection of the zero magnitude and phase plots!!

Gain Plot of Real Pole: $G(s) = \frac{1}{\tau s+1}$

An exact reflection of the zero gain plot

$$20 \log \frac{1}{|i\omega\tau + 1|} = \begin{cases} 0 \text{ dB} & \omega << \frac{1}{\tau} \\ -3.01 \text{ dB} & \omega = \frac{1}{\tau} \\ -20 \log \omega & \omega >> \frac{1}{\tau} \end{cases}$$



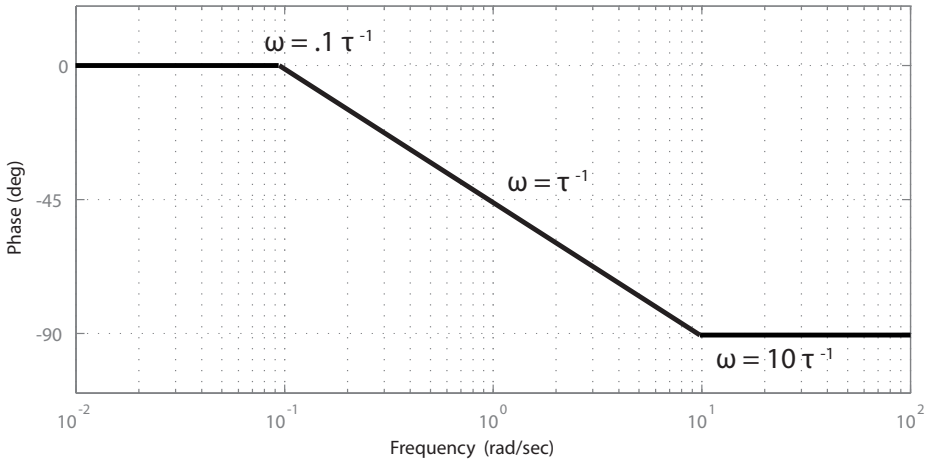


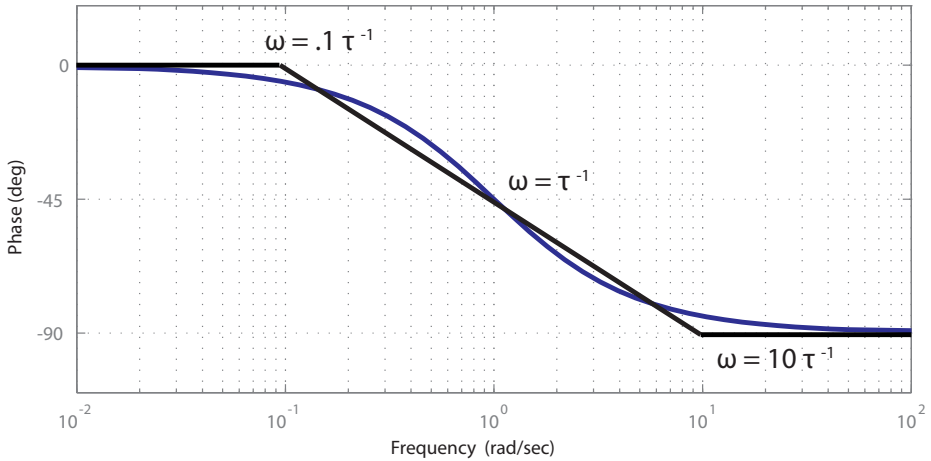
Phase Plot of Real Pole: $G(s) = \frac{1}{\tau s+1}$

Our approximation is also a reflection of the zero

$$\angle(\imath\omega\tau + 1)^{-1}$$

$$= \begin{cases} 0^\circ & \omega < \frac{.1}{\tau} \\ -45^\circ\,(1 + \log\omega\tau) & \frac{.1}{\tau} < \omega < \frac{10}{\tau} \\ -90^\circ & \omega > \frac{10}{\tau} \end{cases}$$





Step 6: Plot Complex Poles and Zeros

Complex Zeros:

$$G_{cz}(s) = \frac{s^2 + 2\zeta\omega_n s + \omega_n^2}{\omega_n^2}$$

Complex Poles:

$$G_{cp}(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{1}{G_{cz}(s)}$$

$$G_{cz}(\imath\omega) = 1 - \left(\frac{\omega}{\omega_n}\right)^2 + 2\zeta\frac{\omega}{\omega_n}\imath = \begin{cases} 1 & \frac{\omega}{\omega_n} << 1 \\ 2\zeta\imath & \frac{\omega}{\omega_n} = 1 \\ -\left(\frac{\omega}{\omega_n}\right)^2 & \frac{\omega}{\omega_n} >> 1 \end{cases}$$

$$20\log|G_{cz}(\imath\omega)|$$

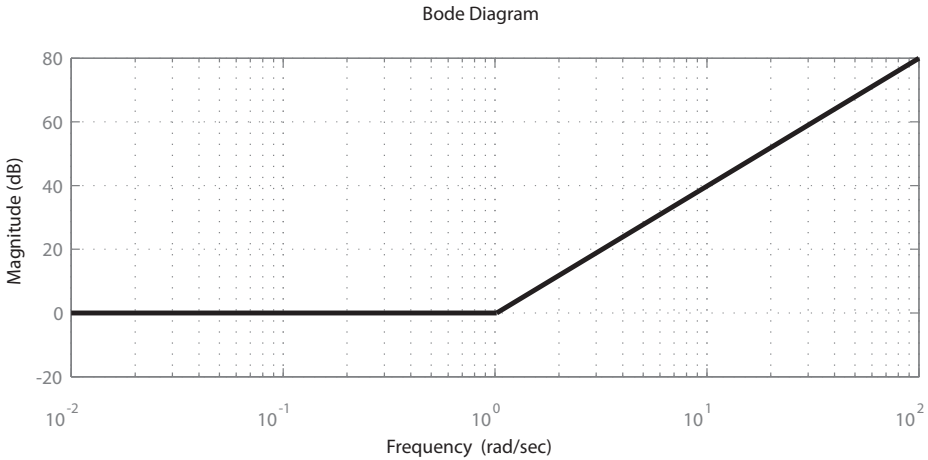
$$= 20\log\left|-\left(\frac{\omega}{\omega_n}\right)^2 + 2\zeta\frac{s\omega}{\omega_n}\imath + 1\right| = \begin{cases} 0dB & \frac{\omega}{\omega_n} << 1 \\ 20\log(2\zeta) & \frac{\omega}{\omega_n} = 1 \\ 40(\log\omega - \log\omega_n) & \frac{\omega}{\omega_n} >> 1 \end{cases}$$

Gain Plot of Complex Poles and Zeros

Complex Zero Gain:

$$20\log|G_1(\imath\omega)| =$$

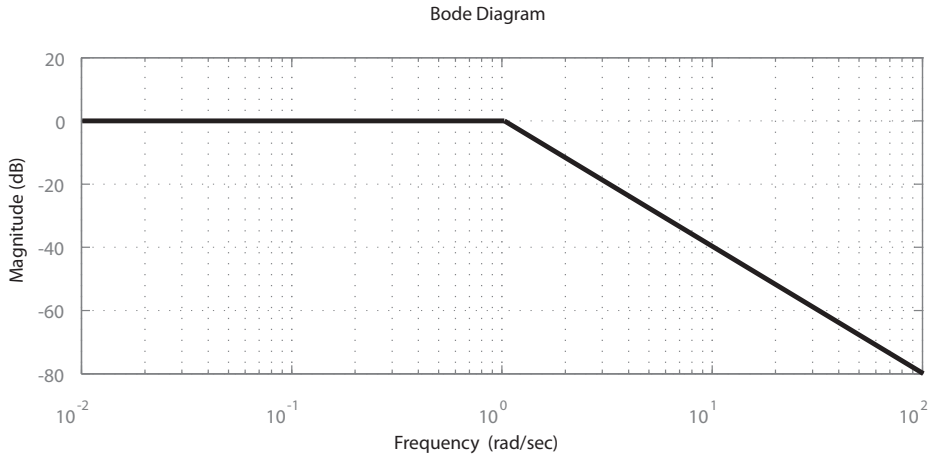
$$= \begin{cases} 0dB & \frac{\omega}{\omega_n} << 1 \\ 20\log(2\zeta) & \frac{\omega}{\omega_n} = 1 \\ 40(\log\omega - \log\omega_n) & \frac{\omega}{\omega_n} >> 1 \end{cases}$$



Complex Pole Gain:

$$20\log|G_1(\imath\omega)|$$

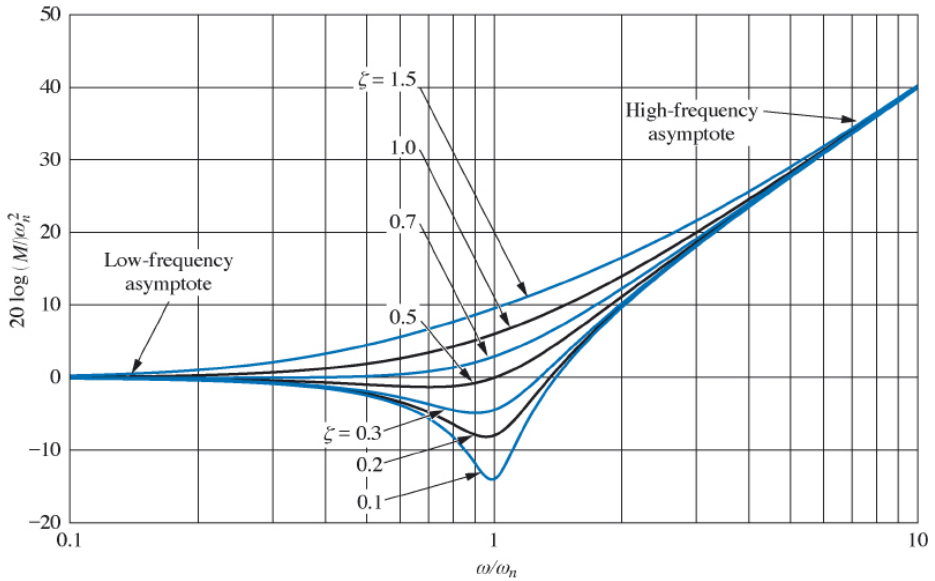
$$= \begin{cases} 0dB & \frac{\omega}{\omega_n} << 1 \\ -20\log(2\zeta) & \frac{\omega}{\omega_n} = 1 \\ -40(\log\omega - \log\omega_n) & \frac{\omega}{\omega_n} >> 1 \end{cases}$$



Gain Behaviour at Natural Frequency for Complex Zeros

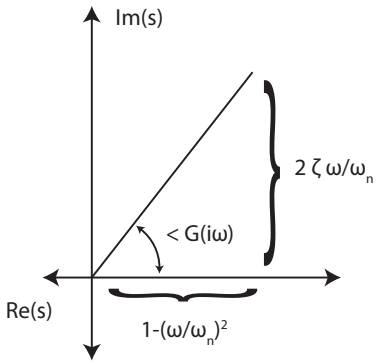
The Behavior near $\omega = \omega_n$ depends on ζ.

- When $\zeta = 0$,
$$20\log(2\zeta) = -\infty$$
- When $\zeta = .5$,
$$20\log(2\zeta) = 0$$
- When $\zeta = 1$,
$$20\log(2\zeta) = 6.02dB$$



Phase Plot of Complex Poles and Zeros

$$\angle G_1(\imath\omega) = \angle\left(1 - \left(\frac{\omega}{\omega_n}\right)^2 + 2\zeta\frac{\omega}{\omega_n}\imath\right) = \tan^{-1}\left(\frac{2\zeta\omega\omega_n}{\omega_n^2 - \omega^2}\right)$$

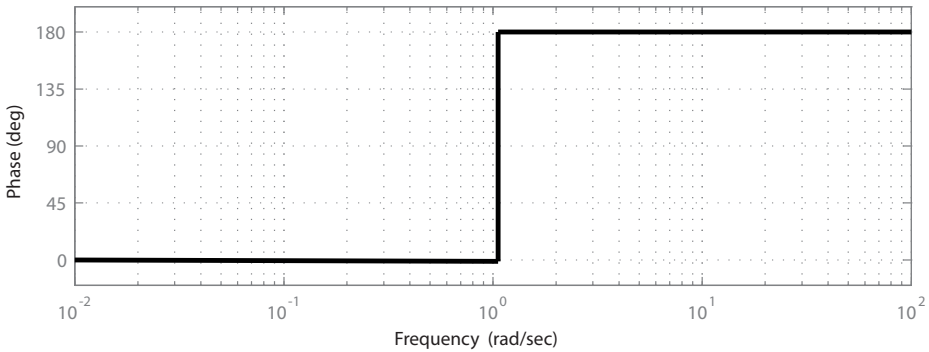


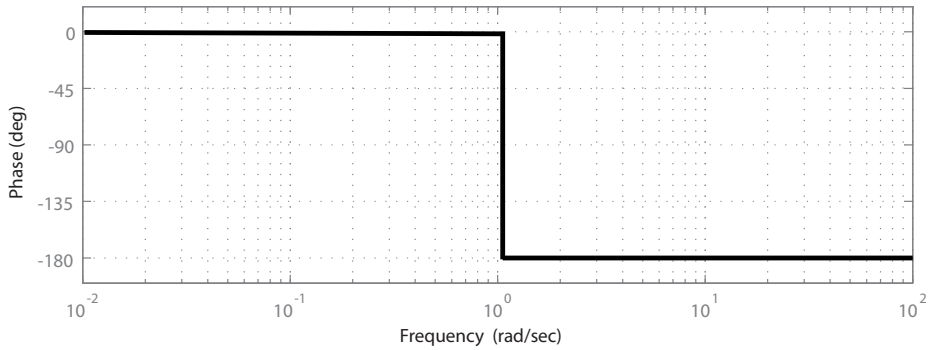
Complex Zero Phase:

$$\angle G_1(\imath\omega) = \begin{cases} 0° & \frac{\omega}{\omega_n} << 1 \\ 90° & \frac{\omega}{\omega_n} = 1 \\ 180° & \frac{\omega}{\omega_n} >> 1 \end{cases}$$

Complex Pole Phase:

$$\angle G_1(\imath\omega) = \begin{cases} 0° & \frac{\omega}{\omega_n} << 1 \\ -90° & \frac{\omega}{\omega_n} = 1 \\ -180° & \frac{\omega}{\omega_n} >> 1 \end{cases}$$



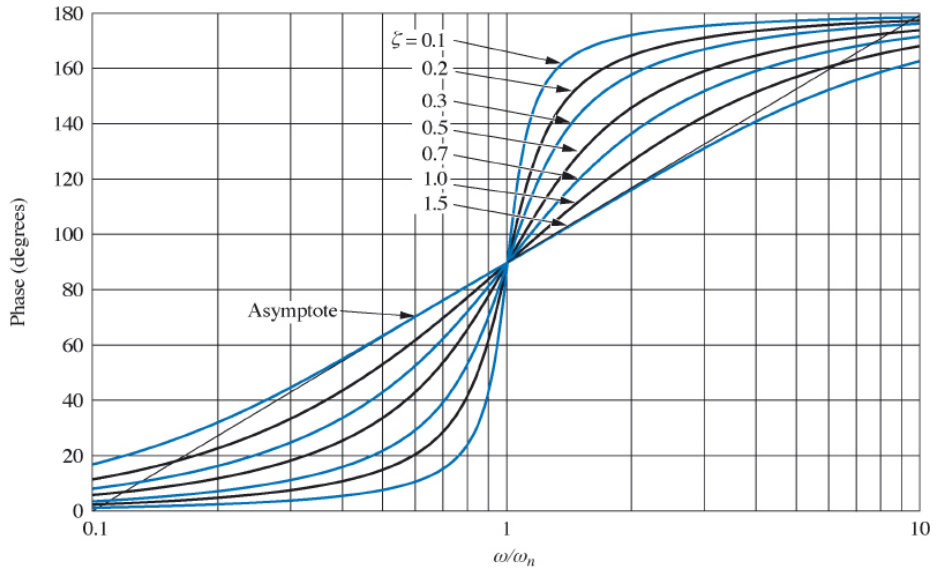


Phase Behaviour at Natural Frequency for Complex Zeros

Slope variation vs. Damping Ratio, ζ, at Natural Frequency

The Behavior near $\omega = \omega_n$ depends on ζ. For $\omega = \omega_n = 1$,

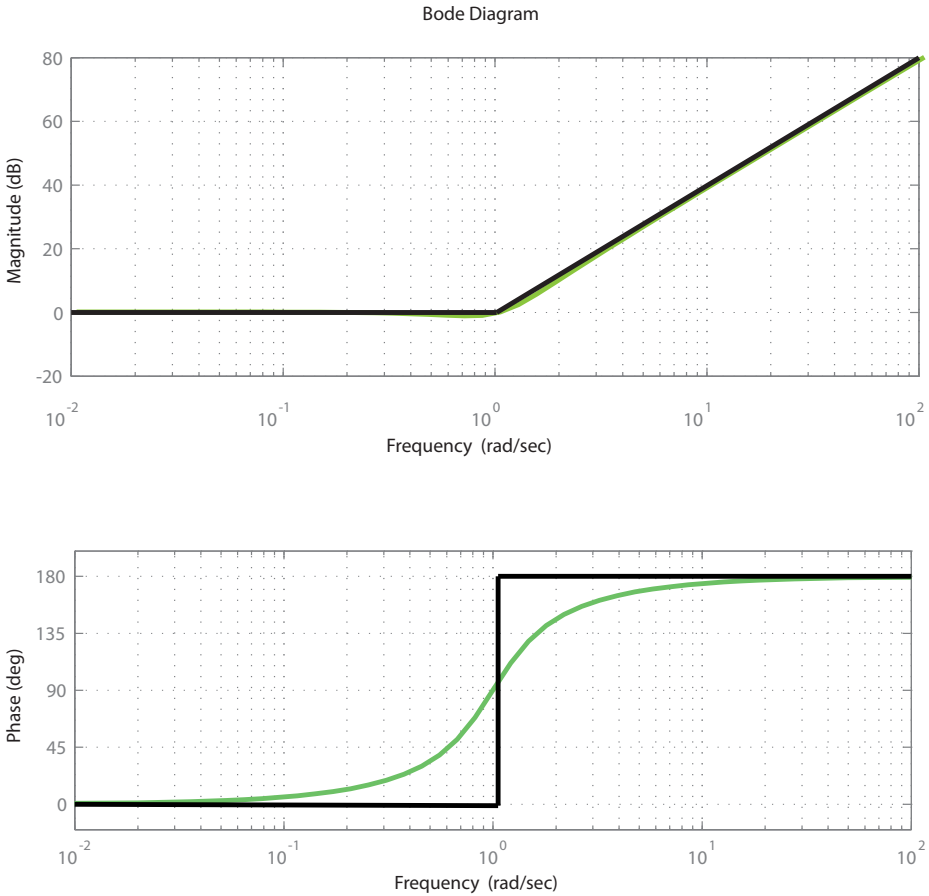
- When $\zeta = 0$, $\dfrac{d\angle G_1(\imath\omega)}{d\omega} = \infty$
- When $\zeta = .1$, $\dfrac{d\angle G_1(\imath\omega)}{d\omega} = 10$
- When $\zeta = .5$, $\dfrac{d\angle G_1(\imath\omega)}{d\omega} = 2$
- When $\zeta = 1$, $\dfrac{d\angle G_1(\imath\omega)}{d\omega} = 1$



Gain and Phase Plots vs. Matlab for Complex Zeros

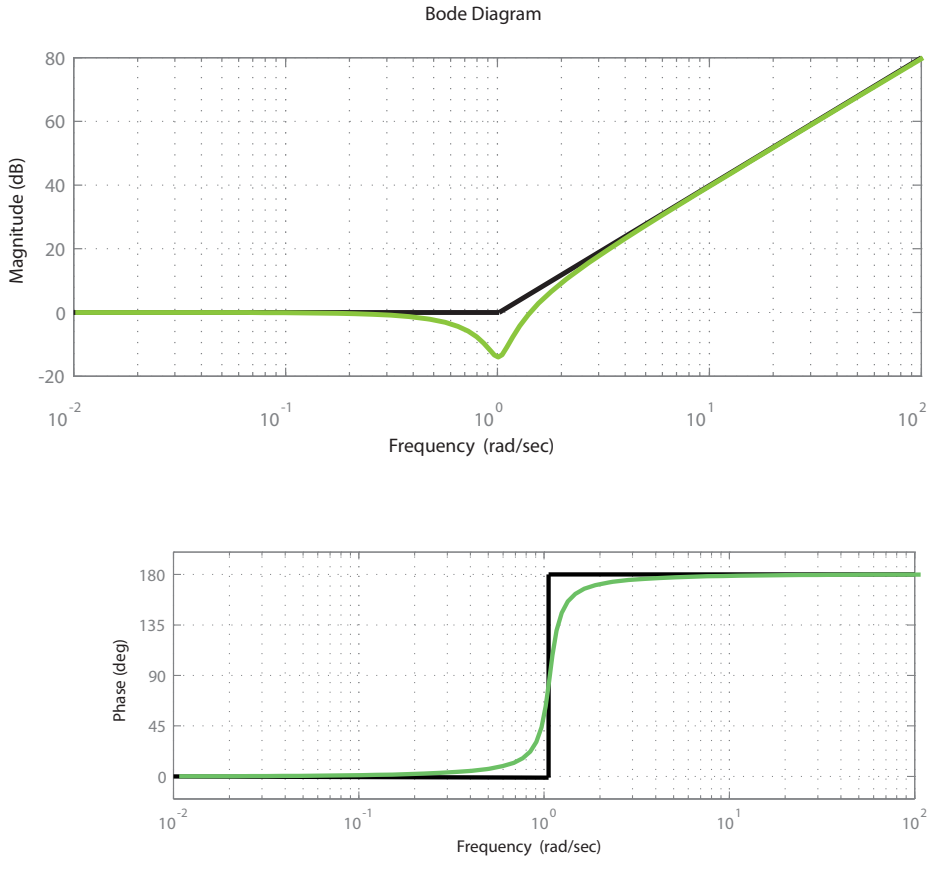
Dependence on Damping Ratio

Comparison for $\zeta = .5$.



Good for magnitude, bad for phase

Comparison for $\zeta = .1$.

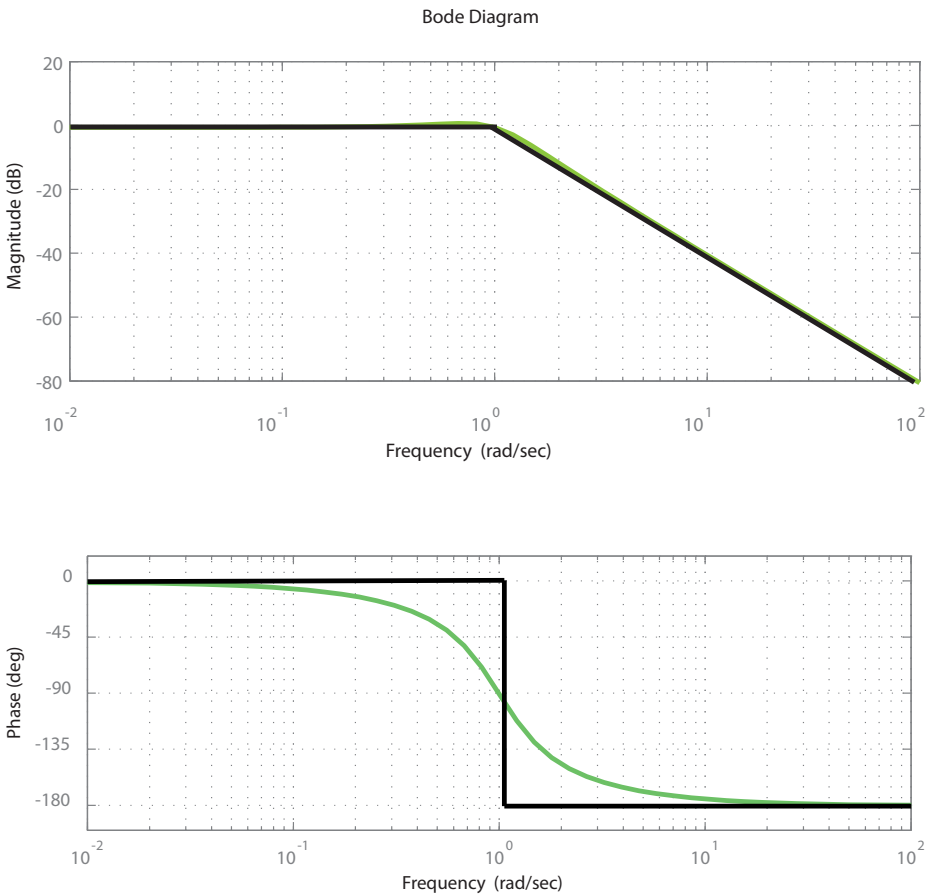


Phase improves, Magnitude gets worse.

Gain and Phase Plots vs. Matlab for Complex Poles

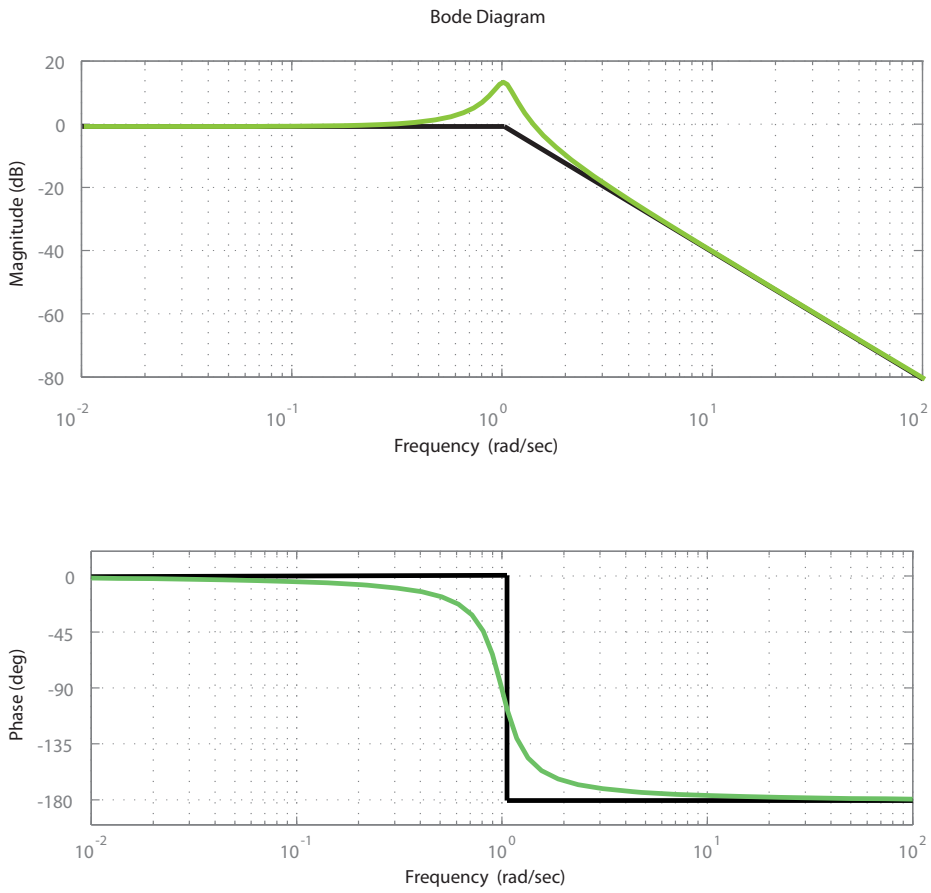
Dependence on Damping Ratio

Comparison for $\zeta = .5$.



Good for magnitude, bad for phase

Comparison for $\zeta = .1$.



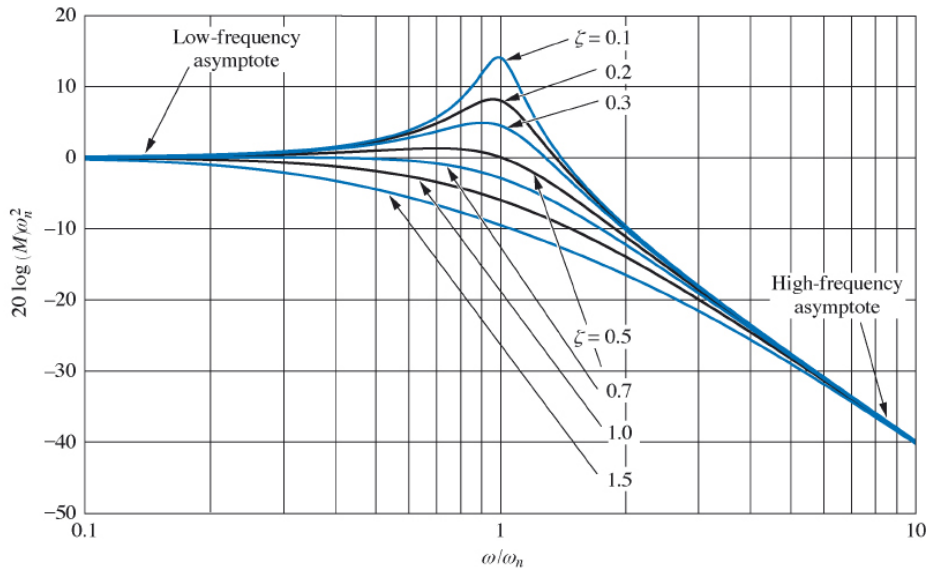
Phase improves, Magnitude gets worse.

Gain Behaviour at Natural Frequency for Complex Poles

The effect of Damping Ratio, ζ

Recall that $|G_0(\imath\omega_n)| = \frac{1}{2\zeta}$.

- When $\zeta = 0$,
$$20\log|G_0(\imath\omega_n)| = 20\log(2\zeta) = +\infty$$
- When $\zeta = .5$,
$$20\log|G_0(\imath\omega_n)| = 20\log 1 = 0$$
- When $\zeta = 1$,
$$20\log|G_0(\imath\omega_n)| = 20\log 2 = -6.02dB$$



Resonance Behaviour at Natural Frequency

Figure: The Tacoma Narrows Bridge

Step 7: Add all the Plots Together

Illustration: Lead Controller

We now have the pieces for a Lead-Lag Compensator.

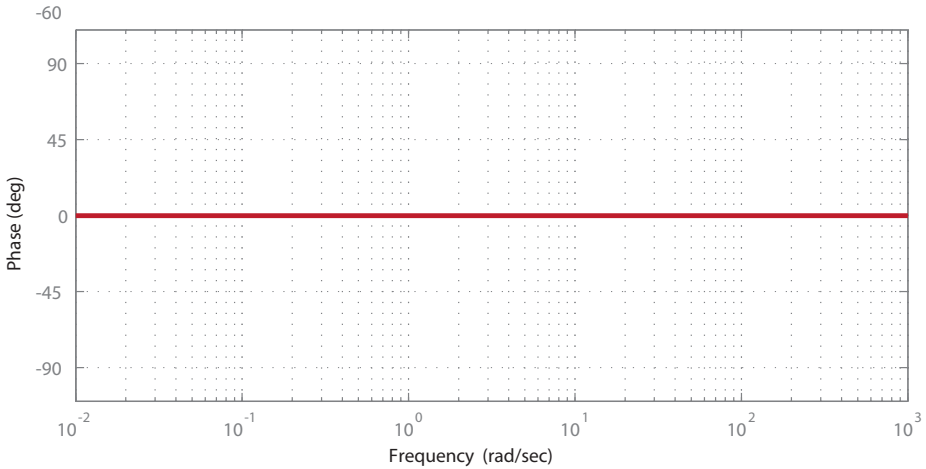
Example: Lead Compensator

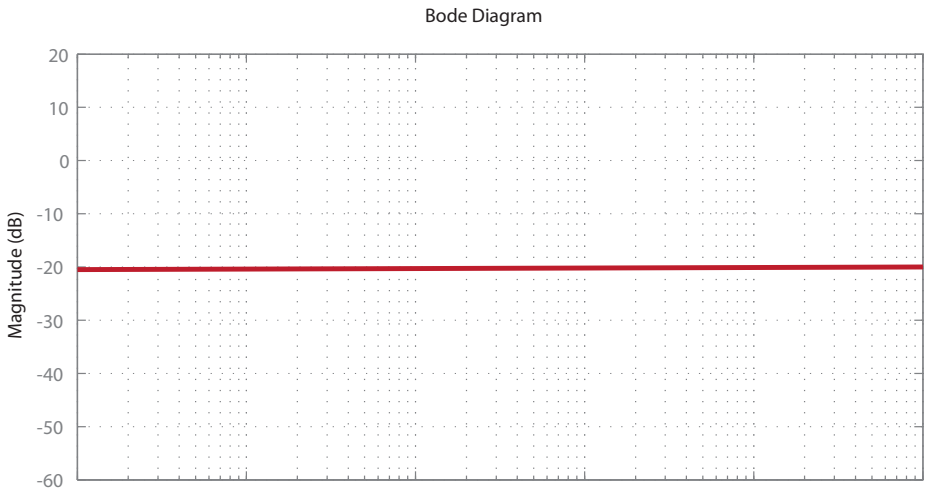
$$G(s) = \frac{s+1}{s+10}$$

First Step: Standard Form

$$G(s) = \frac{s+1}{s+10} = \frac{1}{10}\frac{s+1}{\frac{1}{10}s+1}$$

Second Step: Plot the Constant $c = .1 = -20\text{dB}$.





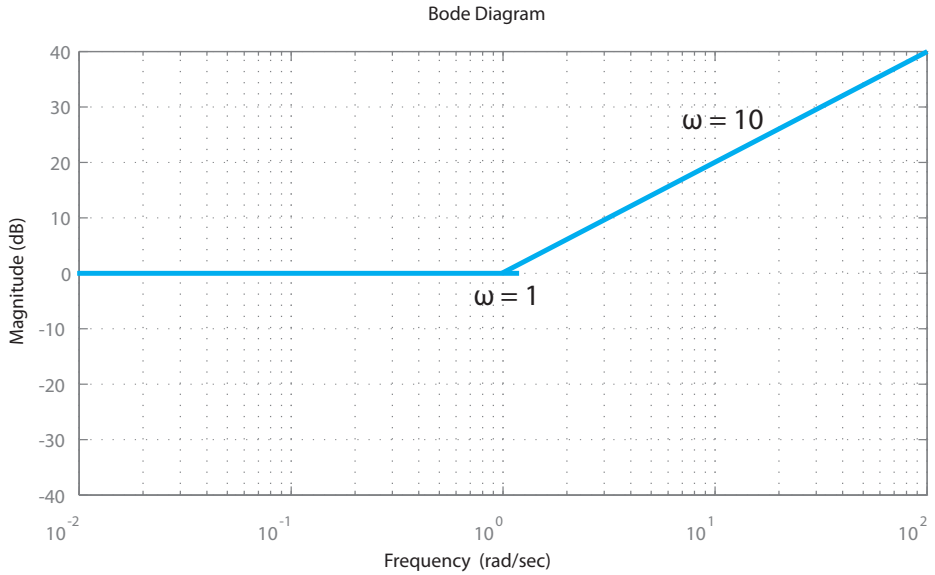
Bode Plot for Lead Compensator

Plot the Zero at $\tau = 1$

Third Step: Plot the Zero for $\tau = 1$.

Magnitude:

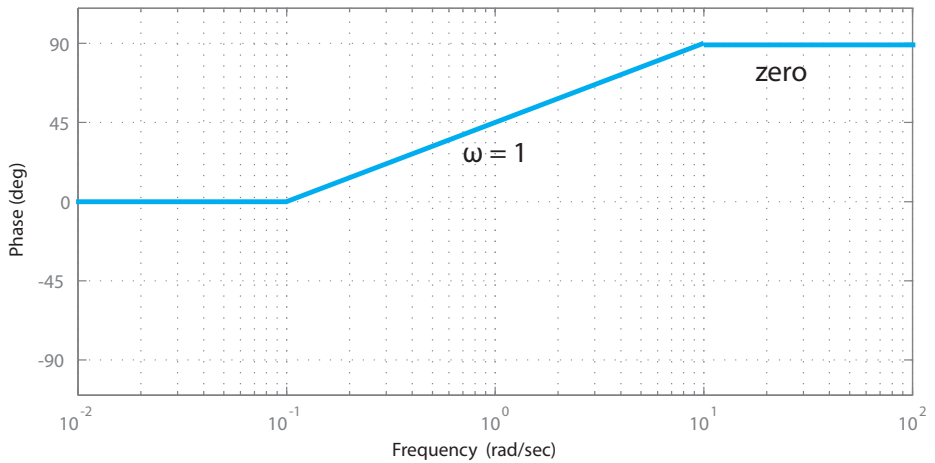
$$20 \log |\imath\omega + 1| = \begin{cases} 0 \text{ dB} & \omega << 1 \\ 3.01 \text{ dB} & \omega = 1 \\ 20 \log \omega & \omega >> 1 \end{cases}$$



Phase:

$$\angle(\imath\omega + 1)$$

$$= \begin{cases} 0^\circ & \omega < .1 \\ 45^\circ \, (1 + \log \omega) & .1 < \omega < 10 \\ 90^\circ & \omega > 10 \end{cases}$$



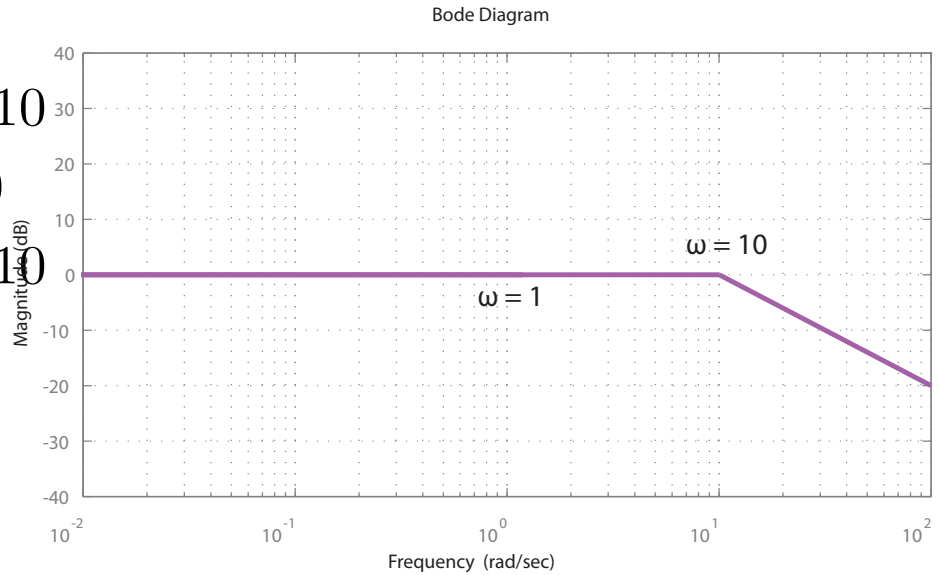
Bode Plot for Lead Compensator

Plot the Pole at $\tau = \frac{1}{10}$

Fourth Step: Plot the Pole at $\omega = \frac{1}{\tau} = 10$.

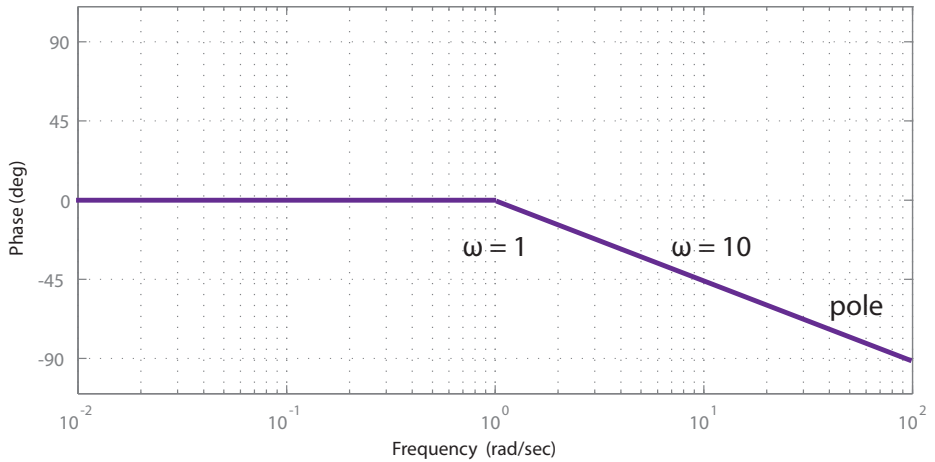
Magnitude:

$$20 \log \left| \frac{\imath\omega}{10} + 1 \right|^{-1} = \begin{cases} 0 \text{ dB} & \omega << 10 \\ -3.01 \text{ dB} & \omega = 10 \\ -20 \log \omega & \omega >> 10 \end{cases}$$



Phase:

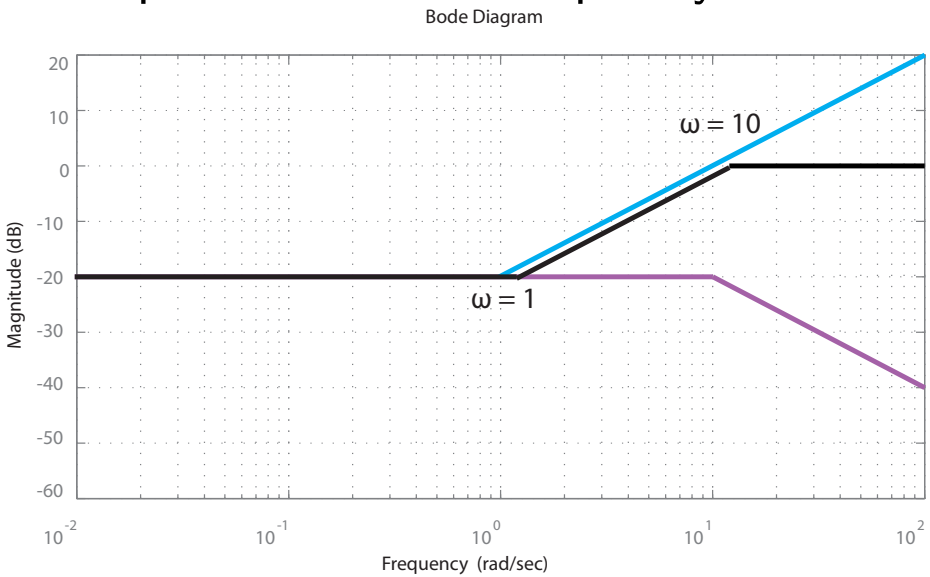
$$\angle \left(\frac{\imath\omega}{10} + 1 \right)^{-1} = \begin{cases} 0^\circ & \omega < 1 \\ -45^\circ \left(1 + \log \frac{\omega}{10} \right) & 1 < \omega < 100 \\ -90^\circ & \omega > 100 \end{cases}$$

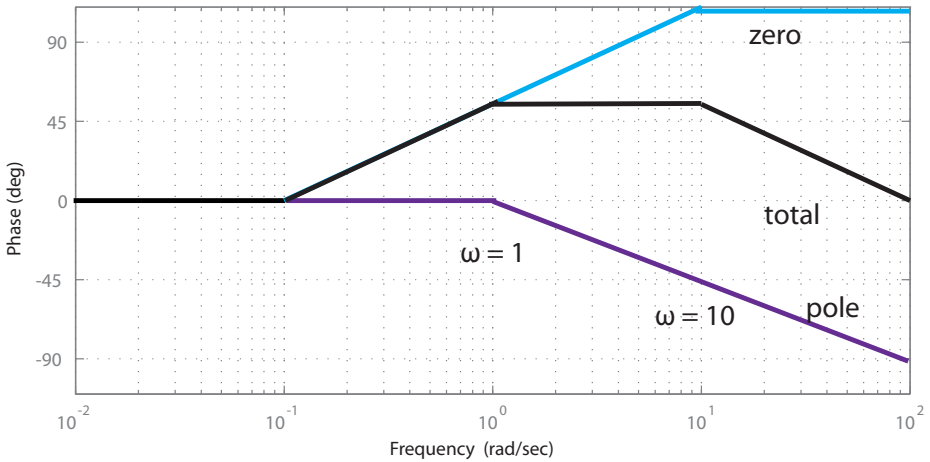


Bode Plot for Lead Compensator

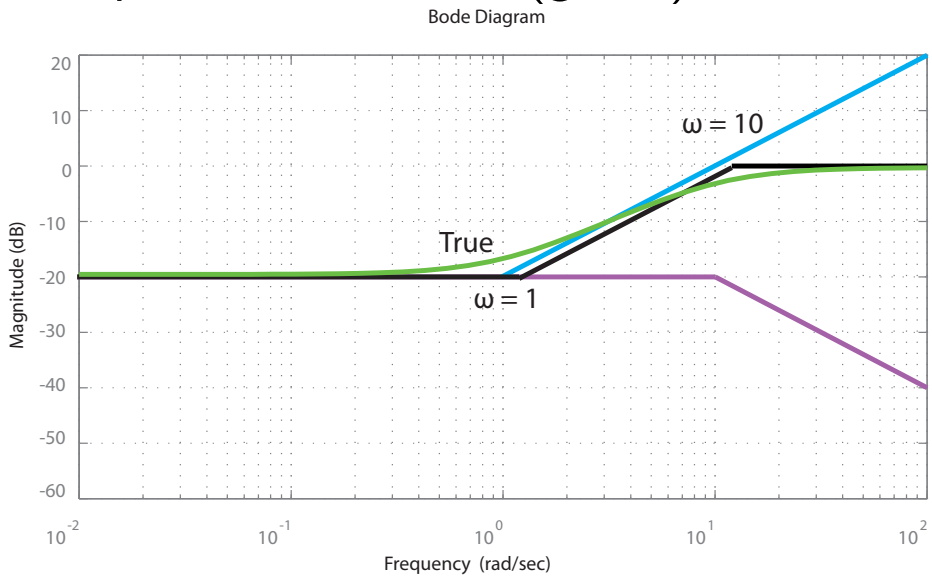
Combine Pole, Zero, and Constant

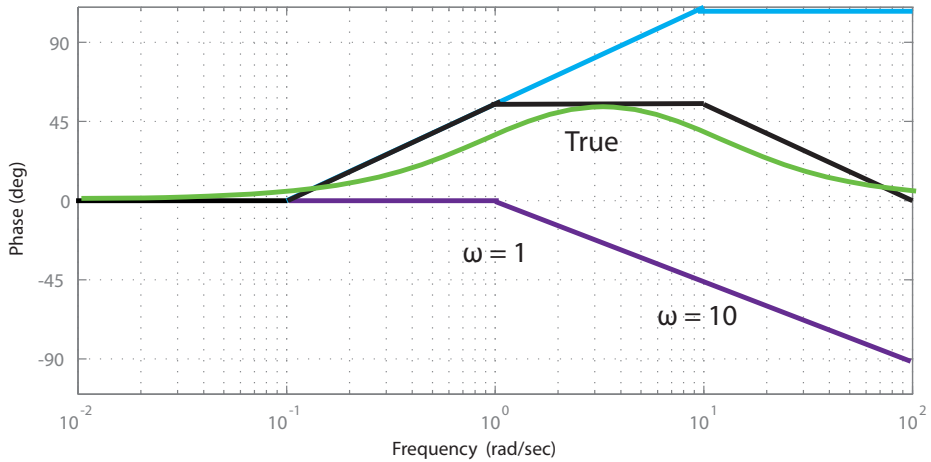
Adds phase for small frequency band.





Compare with Matlab (green)



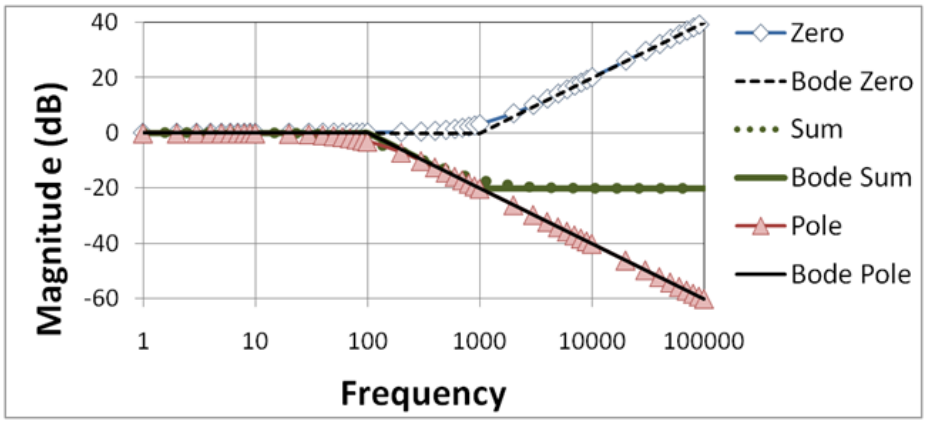


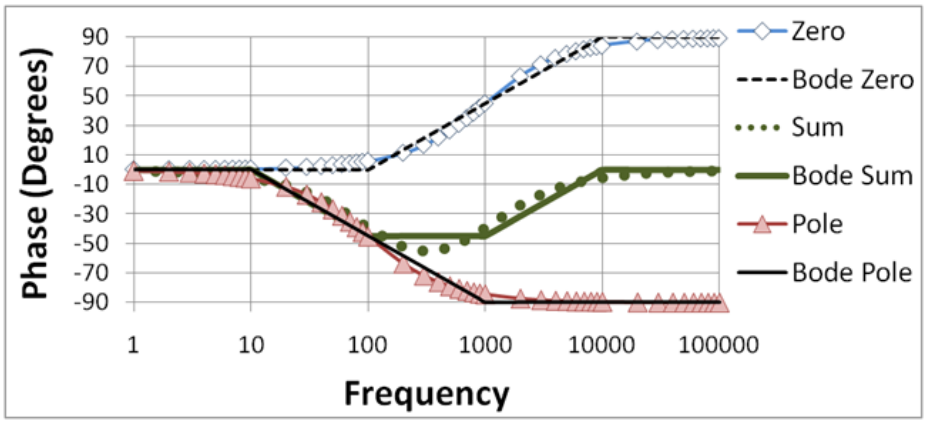
Bode Plot for Lead Compensator

Negative Phase in Narrow Frequency Band

Example: Lag Compensator

$$G(s) = \frac{1}{10}\frac{s+1000}{s+100} = \frac{\frac{s}{1000}+1}{\frac{s}{100}+1}$$





- Pole at $\omega = 100$
- Zero at $\omega = 1000$

Output Phase "Lags" behind the input.

Combine Bode Plots for Lead-Lag Compensator

Good Advice: Tabulate Inflection Points and Slopes

Put them together to get a Lead-Lag Compensator.

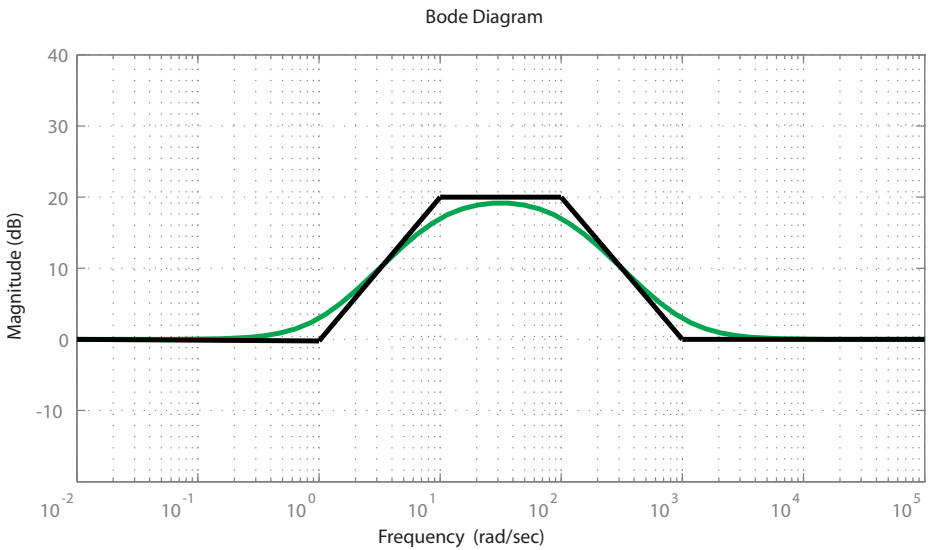
$$G(s) = \frac{s+1}{s+10}\frac{s+1000}{s+100} = \frac{s+1}{\frac{s}{10}+1}\frac{\frac{s}{1000}+1}{\frac{s}{100}+1}$$

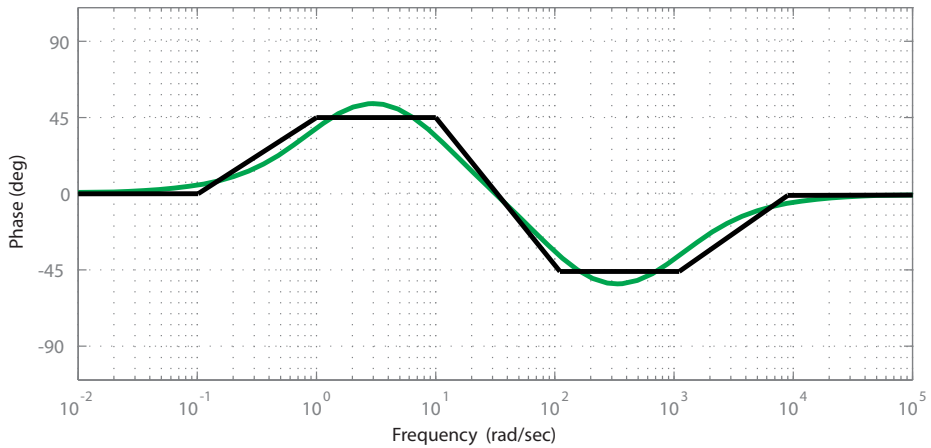
Gain Table: Changes in Slope at Inflection Points

- $\omega = 1$: $+20$ dB/dec
- $\omega = 10$: -20
- $\omega = 100$: -20
- $\omega = 1000$: $+20$

Phase Table: Changes in Slope.

- $\omega = .1$: $+45°$/dec
- $\omega = 1$: $-45°$
- $\omega = 10$: $-90°$
- $\omega = 100$: $+90°$
- $\omega = 1000$: $+45°$
- $\omega = 10000$: $-45°$





Summary

What have we learned today?

Simple Plots

- Real Zeros
- Real Poles
- Complex Zeros
- Complex Poles

Next Lecture: Compensation in the Frequency Domain